

Ranked Video Playlist — Amii × ZeroKey (interview Friday evening)

Compiled: 2026-09-30 · Ranked from `amii-zerokey-video-refresher.md` by importance for this interview: case-study format and the headline motion topic first, then the main topics (ML/stats, time-series DL, computer vision), then minor topics.

One-click playlists

YouTube caps these links at 50 videos, so there are two. Open a link, and YouTube plays the videos in order as a temporary playlist. To keep it in your account, use **Save** (or `:` → *Save to playlist* on the playlist panel) and name it.

- **Playlist 1 — Must-watch before Friday (ranks 1–50):** [open](#)
- **Playlist 2 — Only if time allows (ranks 51–95):** [open](#)

If a link doesn't open as a playlist (YouTube changes this feature from time to time), use the ranked tables below and add videos with *Save* → *New playlist*.

Plan to Friday evening

When	Ranks	Goal
Wednesday evening	1–16	Case format, Kalman, core stats, LSTM/attention, leakage, DTW, the key CV lectures
Thursday	17–35	Rest of the main topics; stop at 35 and rehearse Scenarios T and I out loud
Friday before lunch	36–49	Skim at 1.5–2× speed; skip anything you already know cold
Friday afternoon	50	5-minute Kalman recap, then stop watching and rest

Playlist 1 — Must-watch (ranks 1–50)

#	Video	Why it's here
1	ML System Design Question — Create an ETA System for Maps (full mock)	Case-study format: watch how a strong candidate paces a live design
2	Why Use Kalman Filters? — Understanding Kalman Filters, Part 1	Headline topic: motion analysis starts with Kalman intuition
3	Kalman Filter & EKF	Kalman + EKF equations you may have to write on a whiteboard
4	Machine Learning Fundamentals: Bias and Variance	Most likely opening fundamentals question
5	p-values: What they are and how to interpret them	Statistics is a main topic; p-values are the classic trap
6	ROC and AUC, Clearly Explained!	Evaluation under imbalance (rare risk events)
7	Bayes theorem, the geometry of changing beliefs	Base-rate fallacy: why rare-event alarms are mostly false

#	Video	Why it's here
8	Long Short-Term Memory (LSTM), Clearly Explained	Your core model family (JAES, McGill)
9	Attention in transformers, visually explained	Attention, Q/K/V, $\sqrt{d_k}$
10	Time Series Cross Validation Explained	Leak-free splits: a strong answer most candidates miss
11	Dynamic Time Warping (DTW) Explained	Your DTW → RNN story depends on it
12	Lecture 16: Detection and Segmentation	CV main topic: detection, segmentation, keypoints
13	Lecture 18: Videos	Video/action recognition
14	Realtime Multi-Person 2D Human Pose Estimation using Part Affinity Fields, CVPR 2017 Oral	Pose estimation: the CV topic closest to this role
15	DeepSORT Object Tracking Explained	Tracking = Kalman + matching; links CV to motion
16	ML System Design Mock Interview — Build an ML System That Classifies Which Tweets Are Toxic	Second mock: rare-positive classification
17	Temporal Convolutional Networks, The Next Revolution for Time-Series?	TCNs for streaming edge detection
18	ST-GCN: Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition	Skeleton/graph model closest to multi-tag RTLS
19	Transformer Neural Networks, ChatGPT's foundation, Clearly Explained!!!	Transformer block end to end
20	Precision-Recall Curve Explained with Simple Examples	PR curves for imbalance
21	Bootstrapping Main Ideas!!!	Bootstrap CIs (then block bootstrap for time series)
22	False Discovery Rates, FDR, clearly explained	Multiple comparisons
23	Simpson's Paradox	Confounding; critical-thinking ammunition
24	Causality: Difference-in-Differences	'Did the intervention reduce risk?'
25	Understanding and Avoiding Data Leakage with Hamel Husain	Leakage war stories (Scenario T)
26	Paper review: Towards Total Recall in Industrial Anomaly Detection	Industrial inspection (Xu Han's area, Scenario R)
27	Lecture 15: Object Detection	Detection basics: IoU, NMS, YOLO
28	From FP32 to INT8: Post-Training Quantization Explained in PyTorch	Edge deployment: int8
29	Quaternions and 3d rotation, explained interactively	3D rotations
30	Kabsch-Umeyama Algorithm — How to Align Point Patterns	Aligning trajectories to templates (Scenario P)
31	But what is the Fourier Transform? A visual introduction	Repetitive-motion analysis
32	Change Point Detection in Time Series	
33	Hidden Markov Models (HMMs) Explained with the Viterbi Algorithm	

#	Video	Why it's here
34	MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation (CVPR 2019)	
35	An Image is Worth 16x16 Words (Paper Explained)	
36	Lecture 8: CNN Architectures	
37	Linear Camera Model	
38	Intrinsic and Extrinsic Matrices	
39	Computing Homography	
40	Overview — Optical Flow	
41	Maximum Likelihood, clearly explained!!!	
42	Machine Learning Fundamentals: Cross Validation	
43	Gradient Boost Part 1: Regression Main Ideas	
44	Principal Component Analysis (PCA), Step-by-Step	
45	Probability Calibration for Classification (Platt, isotonic, logistic and beta)	
46	Model Context Protocol (MCP), clearly explained (why it matters)	
47	ReAct AI Agents, clearly explained!	
48	Physics Informed Neural Networks (PINNs)	
49	Knowledge Distillation: A Good Teacher is Patient and Consistent	
50	Kalman Filter — 5 Minutes with Cyrill	

Playlist 2 — If time allows (ranks 51–95)

Mostly deeper versions of Playlist 1 topics, reinforcement learning (not on the recruiter's list), and warm-up basics you likely already know.

#	Video	Why it's here
51	SLAM Course — 06 — Unscented Kalman Filter	
52	MSR Course — 07 Particle Filter	
53	How Does Trilateration Work in GPS Positioning?	
54	The NIOSH Lifting Equation: Example 2	
55	Euler Angles and Gimbal Lock, Explained	
56	Full ML Design Mock by ex-Meta Staff Engineer (with feedback)	
57	Understanding Stationarity, ACF, PACF, and ARIMA Fundamentals	
58	But what is the Central Limit Theorem?	
59	The standard error, Clearly Explained!!!	
60	Confidence Intervals, Clearly Explained!!!	

#	Video	Why it's here
61	In Statistics, Probability is not Likelihood.	
62	Regularization Part 1: Ridge (L2) Regression	
63	Regularization Part 2: Lasso (L1) Regression	
64	Ridge vs Lasso Regression, Visualized!!!	
65	Random Forests Part 1 — Building, Using and Evaluating	
66	XGBoost Part 1: Regression	
67	Singular Value Decomposition (SVD): Overview	
68	A Short Introduction to Entropy, Cross-Entropy and KL-Divergence	
69	Adam Optimization Algorithm (C2W2L08)	
70	Attention for Neural Networks, Clearly Explained!!!	
71	Lecture 12: Recurrent Neural Networks	
72	Lecture 6.1 — Introduction to Graph Neural Networks	
73	Tutorial 17: Self Supervised Contrastive Learning with SimCLR (Part 1)	
74	Chronos: Learning the Language of Time Series	
75	Lecture 7: Convolutional Networks	
76	Gaussian Mixture Model (GMM) for Anomaly Detection	
77	Bayesian Online Change-Point Detection	
78	Neural Ordinary Differential Equations (Neural ODEs): Implementation	
79	DINO: Emerging Properties in Self-Supervised Vision Transformers	
80	Segment Anything Paper Explained	
81	DETR — End to end object detection with transformers (ECCV 2020)	
82	Semantic Segmentation with U Net Made Easy	
83	Lucas-Kanade Method	
84	Dealing with Outliers: RANSAC	
85	Epipolar Geometry — Uncalibrated Stereo	
86	What is Retrieval-Augmented Generation (RAG)?	
87	1hr Talk: Intro to Large Language Models	
88	Low-Rank Adaptation (LoRA) Explained	
89	RL Course by David Silver — Lecture 1: Introduction to Reinforcement Learning	
90	Lecture 21: Reinforcement Learning	
91	Proximal Policy Optimization Explained	
92	Gradient descent, how neural networks learn	
93	Backpropagation, intuitively	

#	Video	Why it's here
94	Backpropagation calculus	
95	But what is a neural network?	